

Breathe Exposure Protocol (BXP) Specification

Version 2.0 — Formal Technical Reference

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Abstract

BXP (Breathe Exposure Protocol) is an open, universal file system specification and data standard for the capture, storage, transmission, and interpretation of atmospheric exposure data. It defines the structural foundation upon which air quality applications, environmental monitoring systems, research databases, health platforms, and community reporting tools can interoperate using a single, standardised, open data layer.

BXP is a pure software protocol standard. Like HTTP, PDF, and MP3 — it requires no proprietary hardware, no centralised infrastructure, and no licensing fees. It runs on any device, any platform, any geography, at any scale.

Implementation Status Legend

Throughout this document, components are labelled as follows:

Label	Meaning
[IMPLEMENTED]	Fully working code exists in this re
[SPECIFIED]	Fully defined in the specification;
[PROPOSED]	Design is outlined; detailed specifi
[FUTURE]	Planned for a future version

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1. Terminology

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in RFC 2119.

Term	Definition
BXP Volume	A complete BXP file system instance
Agent	Any atmospheric substance or biological
Reading	A single point-in-time measurement
Aggregate	A computed summary of multiple readings
Geohash	A compact geographic coordinate encoding
BXP_HRI	BXP Health Risk Index – composite health risk
Container	A portable, self-verifying file bundle
Device Token	Authentication credential issued to a device
Quality Flag	A metadata tag describing data reliability

2. Design Philosophy

BXP is built on six non-negotiable principles:

2.1 Universality [IMPLEMENTED in reference server and SDK]

BXP MUST work for any data source, any platform, any geography, and any scale.

2.2 Interoperability [IMPLEMENTED — JSON format; SPECIFIED — binary format]

Any BXP-compliant system MUST be able to read any BXP-compliant file regardless of origin.

2.3 Openness [IMPLEMENTED]

The standard is fully open. Apache 2.0 licensed. No fees, no closed sections, no gatekeepers. Forever.

2.4 Accessibility [PARTIALLY IMPLEMENTED — Python, TypeScript; FUTURE — MicroPython, Arduino]

BXP MUST be implementable on low-resource devices including basic smartphones and embedded systems.

2.5 Privacy-First [IMPLEMENTED in reference server v2.1]

Individual exposure records are private by default. Cryptographic controls operate at the file level.

2.6 Extensibility [IMPLEMENTED via versioned namespaces]

The standard MUST grow without breaking existing implementations. Backward compatibility is maintained across all MINOR versions.

3. File System Architecture [SPECIFIED — logical structure defined; PROPOSED — physical implementation]

A BXP volume is a logical file system structure. It can be implemented on any physical storage.

3.1 Root Structure

BXP:/

└─ /meta/	– Volume identity, schema version, Merkle checksums
└─ /locations/	– Geohash-based geographic hierarchy
└─ /agents/	– Canonical pollutant definitions
└─ /exposures/	– Device, personal, and aggregate exposure records
└─ /devices/	– Device registry with calibration traceability
└─ /alerts/	– Alert event system
└─ /community/	– Community observation reports
└─ /research/	– Research-grade dataset storage
└─ /system/	– Audit logs, schema version control

Note: The volume file system architecture is defined in the specification. The reference implementation uses a SQLite database rather than a physical BXP volume structure. The volume structure is the target for v3.0 or a dedicated storage implementation.

4. The .bpx File Format

4.1 Overview [JSON: IMPLEMENTED; Binary: SPECIFIED]

Two representations exist, semantically identical and losslessly convertible:

- * Binary .bpx — compact format with 32-byte header, designed for constrained devices
- * JSON .bpx.json — human-readable format, designed for APIs and developer use

4.2 Binary Header (32 bytes) [SPECIFIED]

Offset	Length	Field	Description
0x00	4 bytes	Magic Number	0x42585000 ("BXP\0")
0x04	2 bytes	Major Version	Schema major version (uint16 big-end)
0x06	2 bytes	Minor Version	Schema minor version (uint16 big-end)
0x08	1 byte	File Type	0x01=reading, 0x02=aggregate, 0x03=a
0x09	1 byte	Flags	bit0=compressed, bit1=encrypted, bit
0x0A	2 bytes	Reserved	Must be 0x0000 in v2.0
0x0C	8 bytes	Timestamp	Unix epoch microseconds (int64 big-e
0x14	4 bytes	Payload Length	Payload size in bytes (uint32 big-en
0x18	4 bytes	Header Checksum	CRC32 of bytes 0x00–0x17
0x1C	4 bytes	Payload Checksum	CRC32 of entire payload

4.3 Reading Record Schema (JSON) [IMPLEMENTED]

```
{  
  "bpxVersion": "2.0",
```

```

"fileType": "reading",
"deviceUuid": "550e8400-e29b-41d4-a716-446655440000",
"geohash": "s1v0g",
"latitude": 5.5571,
"longitude": -0.1969,
"timestampUs": 1710000000000000,
"durationS": 60,
"indoorOutdoor": "outdoor",
"agents": [
  {
    "agentId": "PM2_5",
    "value": 47.3,
    "unit": "ug/m3",
    "uncertainty": 3.1,
    "method": "optical",
    "belowLod": false
  }
],
"context": {
  "temperatureC": 28.4,
  "humidityPct": 72.1,
  "pressureHpa": 1012.3
},
"quality": {
  "flag": "UNVALIDATED",
  "confidence": 0.9,
  "qcMethod": "bxp-sdk-auto",
  "notes": null
},
"payloadHash": "sha256:..."
}

```

4.4 Required Fields for a Valid Reading [IMPLEMENTED — validated in SDK and server]

Field	Requirement	Notes
bxpVersion	REQUIRED	Must be "2.0"
deviceUuid	REQUIRED	UUID v4
geohash	REQUIRED	Minimum precision 5
timestampUs	REQUIRED	Unix epoch microseconds, UTC
agents	REQUIRED	At least one entry with agentId and
quality.flag	REQUIRED	One of: VALIDATED, UNVALIDATED, SUSP

4.5 Quality Flags [IMPLEMENTED]

Flag	Description
VALIDATED	Passed all automated QC checks
UNVALIDATED	Raw data, no QC applied
SUSPECT	Failed one or more QC checks
INVALID	Known bad data – do not use

4.6 Verification Process [IMPLEMENTED in SDK and server]

1. Canonically serialize payload using sorted JSON keys, compact separators
2. Compute SHA-256 hash of the UTF-8 encoded serialization
3. Prefix with "sha256:" and store as payloadHash
4. On verification: recompute and compare against stored hash

5. If Ed25519 signature present: verify against device public key

5. Agent Schema [IMPLEMENTED — schema defined; thresholds implemented in SDK and server]

5.1 Particulate Matter

Agent ID	Name	Unit	WHO Annual	WHO 24h
PM1	Particulate Matter <1µm	µg/m ³	—	—
PM2_5	Fine Particulate Matter <2.5µm	µg/m ³	5	15
PM10	Coarse Particulate Matter <10µm	µg/m ³	15	45
BC	Black Carbon	µg/m ³	—	—

5.2 Gaseous Pollutants

Agent ID	Name	Unit	WHO Limit
C0	Carbon Monoxide	ppm	4 mg/m ³ (24h)
C02	Carbon Dioxide	ppm	1000 (indoor guideline)
N02	Nitrogen Dioxide	ppb	10 µg/m ³ annual
S02	Sulphur Dioxide	ppb	40 µg/m ³ (24h)
O3	Ground-level Ozone	ppb	100 µg/m ³ (8h)
H2S	Hydrogen Sulphide	ppb	7 ppb
NH3	Ammonia	ppm	25 ppm

5.3 Volatile Organic Compounds

Agent ID	Compound	Unit	Threshold	Classification
TVOC	Total VOC	ppb	500	Aggregate
BENZ	Benzene	ppb	1	Group 1 Carcinogen (IARC)
FORM	Formaldehyde	ppb	8	Group 1 Carcinogen (IARC)
TOLU	Toluene	ppm	50	Neurotoxin

5.4 Biological, Heavy Metal, and Environmental Agents

See Appendix A for the complete 31-agent reference.

5.5 BXP_HRI (Derived Index) [IMPLEMENTED]

A composite 0–100 health risk score. See Section 10 for full specification.

6. Protocol Stages [IMPLEMENTED — Stages 1-4 implemented in reference server]

Stage 1 — LOCATE

Every BXP record MUST be geographically contextualised. Minimum geohash precision: 5.

Stage 2 — DETECT

BXP accepts three tiers of data sources: Tier 1 (phone, manual), Tier 2 (consumer sensors, satellite), Tier 3 (reference instruments).

Stage 3 — INTERPRET

Automated QC rules: range validation, spike detection, flatline detection, humidity correction

(Barkjohn correction for PM2.5 at RH >75%), cross-sensor consistency.

Stage 4 — PROTECT

Six-level risk framework using BXP_HRI. See Section 10.

Stage 5 — REPORT [IMPLEMENTED in reference server v2.1]

Community report schema combining sensor readings with human observations.

7. REST API Specification [IMPLEMENTED — reference server v2.1]

7.1 Base URL

```
https://[host]/bxp/v2/
```

7.2 Implemented Endpoints

Method	Endpoint	Description
GET	/	Interactive dashboard
GET	/bxp/v2/health	Node health and status
GET	/bxp/v2/city/{city}	Live city data via AQICN
GET	/bxp/v2/readings	List readings (with filters)
POST	/bxp/v2/readings	Submit readings (returns 201)
GET	/bxp/v2/readings/{id}	Get reading by ID
DELETE	/bxp/v2/readings/{id}	Delete with cryptographic proof
GET	/bxp/v2/readings/{id}/verify	Integrity check
GET	/bxp/v2/locations/{geohash}/latest	Latest reading for location
GET	/bxp/v2/locations/{geohash}/history	Historical readings
GET	/bxp/v2/locations/{geohash}/aggregat	k≥5 anonymised aggregate
POST	/bxp/v2/devices/register	Register device
GET	/bxp/v2/devices/{uuid}	Get device metadata
POST	/bxp/v2/community/reports	Submit community report
GET	/bxp/v2/community/reports	Get community reports
GET	/bxp/v2/search	Search by city or coordinates
GET	/bxp/v2/nodes	Federated node registry
GET	/metrics	Prometheus format metrics
GET	/bxp/v2/widget/{city}	Embeddable iframe widget

7.3 Standard Response Envelope

```
{
  "status": "ok",
  "bxpVersion": "2.0",
  "data": {},
  "errors": []
}
```

7.4 Authentication

Bearer token authentication. Three token classes: Device Token (per-UUID, permanent), User Token (personal records, 30-day), API Key (administrative). Anonymous submissions are accepted but marked.

7.5 Rate Limiting [IMPLEMENTED]

- * Submissions: 30 per minute per IP
 - * City lookups: 60 per minute per IP
 - * Device registrations: 5 per minute per IP
-

8. Security & Privacy Framework [IMPLEMENTED in reference server v2.1]

8.1 Privacy Rules (Non-Negotiable)

- * Personal exposure records are private by default
- * Person identifiers are NEVER stored in plain form — only SHA-256 hashes
- * Location precision for personal records defaults to Geohash 5
- * Community aggregates are k-anonymised before publication (minimum k=5)
- * Users can permanently delete all personal records via single API call
- * Deletion is cryptographically verifiable and irreversible (deletion proof returned)

8.2 Encryption Standards [SPECIFIED — design defined; not all implemented in reference server]

Data Type	Encryption	Status
Personal records at rest	AES-256-GCM	Specified
Data in transit	TLS 1.3	Specified (deployment-dependent)
Device signatures	Ed25519	Specified
Volume integrity	SHA-256 Merkle tree	Specified
Payload integrity	SHA-256	Implemented

8.3 Data Integrity Model [IMPLEMENTED — Layer 2 record-level hashing]

Layer 1 (file-level CRC32) requires binary format. Layer 2 (SHA-256 per record) is implemented. Layers 3–4 (volume Merkle tree, audit log) are specified for future implementation.

8.4 Compliance Framework [DESIGNED — not formally audited]

BXP is designed for compliance with GDPR, CCPA, POPIA, PDPA, and HIPAA through its privacy architecture. Formal compliance assessment has not been conducted.

9. Community Reporting Layer [IMPLEMENTED in reference server v2.1]

Community reports enable human observers to submit structured qualitative observations alongside optional sensor data. The report schema includes: location, timestamp, visual observations (smoke, haze, visibility), olfactory observations, symptom reports, 12 standardised event tags (open_burning, harmattan, wildfire, etc.), photo attachment metadata, and free-text description.

A five-stage automated QC pipeline processes community reports: geospatial clustering, cross-validation with sensor readings, outlier detection, reporter reliability scoring, and abuse detection.

10. BXP Health Risk Index (BXP_HRI) [IMPLEMENTED]

10.1 Agent Weights

Agent	Weight	Basis
PM2.5	0.35	Highest global DALY burden (WHO 2021)
PM10	0.15	Significant respiratory burden
NO2	0.15	High urban prevalence
O3	0.12	Respiratory effects, summer peaks
CO	0.10	Acute toxicity
SO2	0.05	Industrial variation
TVOC	0.04	Indoor carcinogen concern

10.2 Calculation

```
agent_risk(i) = min(1.0, value(i) / WHO_threshold(i))
raw_HRI       = Σ (agent_risk(i) × weight(i))
BXP_HRI       = min(100, raw_HRI × 100 × duration_factor × vulnerability_factor)
```

Duration factors: 1h → 1.0, 8h → 1.2, 24h → 1.5

Vulnerability factors: general → 1.0, sensitive → 1.3

10.3 Risk Levels

Level	BXP_HRI	Color	General Population Guidance
CLEAN	0–20	#00C851	No restrictions
MODERATE	21–40	#FFBB33	Sensitive groups: limit heavy exerti
ELEVATED	41–60	#FF8800	Reduce outdoor exertion
HIGH	61–75	#CC0000	N95 outdoors, close windows
VERY_HIGH	76–90	#9B0000	Avoid outdoor activity
HAZARDOUS	91–100	#4A0000	Health emergency

11. Governance & Versioning

11.1 BXP Foundation [PROPOSED — does not currently exist as a formal organisation]

The specification proposes governance by an independent, non-profit open standards body — the BXP Foundation — modelled on established open standards bodies such as the IETF and W3C. This governance structure is an aspiration for the protocol's long-term management. It does not currently exist.

11.2 Versioning Policy [IMPLEMENTED]

BXP uses semantic versioning: MAJOR.MINOR.PATCH. MAJOR changes require 18 months advance notice. MINOR changes are backward compatible. All changes go through a public RFC process.

12. Compatibility Matrix

Standard	BXP Compatibility	Status
US AQI	BXP_HRI maps to AQI_US derived field	Designed

WHO AQG 2021 | All thresholds aligned | Implemented
HL7 FHIR R4 | Exposure records map to FHIR Observations | Designed, not documented
OGC SensorThings | Compatible observation model | Designed
GDPR/CCPA | Privacy framework designed for compliance | Designed, not audited

Appendix A — Complete Agent Reference

Agent ID	Full Name	Category	Unit	WHO Limit
PM1	Particulate Matter <1µm	Particulates	µg/m³	—
PM2_5	Fine Particulate Matter	Particulates	µg/m³	5 annual
PM10	Coarse Particulate Matter	Particulates	µg/m³	15 annual
BC	Black Carbon	Particulates	µg/m³	—
CO	Carbon Monoxide	Gas	ppm	4 mg/m³ 24h
CO2	Carbon Dioxide	Gas	ppm	1000 indoor
NO2	Nitrogen Dioxide	Gas	ppb	10 µg/m³ annual
NO	Nitric Oxide	Gas	ppb	—
SO2	Sulphur Dioxide	Gas	ppb	40 µg/m³ 24h
O3	Ground-level Ozone	Gas	ppb	100 µg/m³ 8h
H2S	Hydrogen Sulphide	Gas	ppb	7 ppb
NH3	Ammonia	Gas	ppm	25 ppm
TVOC	Total VOC	VOC	ppb	500 ppb
BENZ	Benzene	VOC	ppb	No safe level (IARC Group 1)
FORM	Formaldehyde	VOC	ppb	8 ppb
TOLU	Toluene	VOC	ppm	50 ppm
XYLE	Xylene	VOC	ppm	100 ppm
NAPH	Naphthalene	VOC	ppb	9.4 ppb
MOLD_S	Mold Spores	Biological	spores/m³	500 indoor
POLL_G	Grass Pollen	Biological	grains/m³	50 moderate
POLL_T	Tree Pollen	Biological	grains/m³	100 high
BACT_T	Total Bacteria	Biological	CFU/m³	500 indoor
DUST_M	Dust Mite Allergen	Biological	ng/m³	2 ng/m³
PB	Lead	Heavy Metal	µg/m³	0.5 annual
HG	Mercury	Heavy Metal	µg/m³	1 annual
AS	Arsenic	Heavy Metal	ng/m³	6.6 annual
TEMP	Temperature	Environmental	°C	—
RH	Relative Humidity	Environmental	%	—
PRESS	Atmospheric Pressure	Environmental	hPa	—
UV	UV Index	Environmental	index	—
BXP_HRI	BXP Health Risk Index	Derived	0–100	—

Appendix B — Geohash Reference

Precision	Cell Width	Cell Height	BXP Use
5 chars	~4.9 km	~4.9 km	Minimum valid BXP reading
6 chars	~1.2 km	~0.61 km	Neighbourhood level
7 chars	~153 m	~153 m	Recommended for fixed sources
8 chars	~38 m	~19 m	Street level
9 chars	~4.8 m	~4.8 m	High-resolution personal sensors

Appendix C — Error Codes

Code	HTTP	Meaning
BXP_4001	400	Geohash precision below minimum (5)
BXP_4002	400	Missing required field
BXP_4003	400	Invalid agent ID

BXP_4004		400		Unit not recognised for agent
BXP_4005		400		Timestamp outside acceptable range
BXP_4006		400		Payload hash mismatch
BXP_4010		401		Missing or invalid authentication to
BXP_4290		429		Rate limit exceeded
BXP_5000		500		Internal server error
BXP_5030		503		BXP volume temporarily unavailable

Appendix D — Glossary

Term		Definition
Agent		Any atmospheric substance or biological
BXP Volume		A complete BXP file system instance
Container		A portable self-verifying file bundle
Geohash		A compact geographic coordinate encoding
BXP_HRI		BXP Health Risk Index – composite health
Reading		A single point-in-time measurement
Quality Flag		Metadata tag: VALIDATED / UNVALIDATED
k-anonymisation		No aggregate published with fewer than k
Tier 1/2/3		Source capability tiers: Tier 1 phone, Tier 2
DALY		Disability-Adjusted Life Year – WHO

Origin & Authorship

BXP was conceived and first implemented by Elvarin on February 15, 2026. This document is the official BXP v2.0 Technical Specification.

References

World Health Organization. (2021). WHO Global Air Quality Guidelines. Geneva: WHO.
 Simcoe, T. (2012). Standard setting committees. American Economic Review, 102(1), 305–336.

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